BOE-705	Queuing Theory and Modelling	L T P 2 1 0	3 Credits

Pre-requisites: None

Course Objectives:

CO1	To provide the ideas of formulating mathematical modeling
CO2	To give students a firm foundation in the advanced optimization techniques
CO3	Introduction to Queuing Systems and Notation.
CO4	Understand and compute quantitative metrics of performance for queuing systems.
CO5	To learn about maintenance and replacement systems

Detailed Syllabus

MODULE I

Systems, modelling, general systems theory, concept of simulation, simulation as a decision making tool, types of simulation. Pseudo random numbers, methods of generating random variables, discrete and continuous distributions, testing of random numbers, concepts of Queuing theory, The simplex computational procedure, development of minimum feasible solution, a first feasible solution using slack variables, the artificial basis technique. Two phase method and Charnels Method with artificial variables.

MODULE II

The duality problem of linear programming and its economic interpretation, transportation and assignment problems. Queueing Theory: Introduction of the queuing system, Various components of a queueing system. Pure Birth Process; Pure Death Process, Birth and Death Process, M/M/1, M/M/1 (Generalized), M/M/1/FCFS/K/∞, M/M/C, Erlang's loss model.

MODULE III

Case studies: Development of simulation models using simulation language studied for systems like queuing systems, Production systems, Inventory systems, maintenance and replacement systems and Investment analysis.

Course Outcomes: After the completion of the course the student will be able to:

CO1	Design of experts system.
CO2	Study to design a model and simulate it for any real life complex problem.
CO3	Study of queuing model theory that is used to model and simulate real life problem.
CO4	Study to design a model and simulate it for any real life complex problem.
CO5	Explain about maintenance and replacement systems

Text Books:

1. Hillier, F.S. and Liebermann, G.J. (2009): Introduction to Operations Research; 9th Edition, McGraw Hill.
2. Gass, S.I. (2010): Linear Programming, Methods and Applications, 5th Edition, Dover Books.